UNSW COMP9900 Information Technology Project

Project Proposal

Project Information

Component	Information
Course Code and Course Title	COMP9900 - Information Technology Project
Project Number and Title	Project 26 AI Mediated Teacher Decision Support Tool for Data Informed Lesson Planning
Team Name	F15C-BREAD
Project Client	Sara Mashayekh
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1. Background

1.1 Problem Definition and Impact

Problem Domain Data-informed teaching method has become an important part of modern educational practice (Mandinach & Gummer, 2016). The Australian Professional Standards for Teachers also highlight that teachers should use assessment data to understand students' learning needs, improve their teaching, and help students progress (AITSL, 2011).

Ideally, teachers should use data from various sources to plan courses. However, there are some Challenges.

Current Challenges

- **Inability to Interpret Data Patterns:** Identifying common misconceptions or recognising differences between students based on collected data rather than communicating directly with students is not an easy task for pre-service and novice teachers. Understanding the data itself is already a challenge, let alone translating that information into appropriate instructional decisions (Mandinach & Gummer, 2016).
- **Pedagogical Experience Gap:** A single indicator, such as a low score on a specific question, may correspond to multiple underlying causes, like a student's knowledge gap or a poorly designed assessment. Pre-service teachers often feel confused when interpreting these indicators because they lack the necessary pedagogical experience to connect data with teaching strategies.
- **Invisible Reasoning in Planning:** Pre-service teachers often lack effective support in turning student data into evidence-based lesson planning decisions, primarily because the expert reasoning required for this process remains invisible to them.
- **Black-Box AI System Constraints:** Existing generative AI tools appear to have the potential to become an effective tool since they can quickly generate lesson plans and teaching materials (UNESCO, 2023). However, many of these tools focus on producing outputs rather than helping teachers understand the reasoning behind educational decisions (UNESCO, 2023).
- **High Administrative Overload:** Teachers often work under significant time and workload pressures. According to the OECD TALIS 2024 report, around 52% of teachers identify administrative work as a source of work-related stress, while 37% report that modifying lessons for students with special education needs is a source of stress (OECD, 2025).

Impact

- **Compromised Critical Analysis Time:** From the perspective of pre-service teachers, generative AI tools do not seem to effectively solve their dilemma, and may even make the situation worse because when they receive a complete lesson plan from generative AI tools, they may not have enough time to explore 'why' but to follow AI's plan.

- **Over-Dependence and Deskilling:** If unaddressed, pre-service teachers and novice teachers may become overly dependent on AI-generated content, failing to develop the professional judgement required to evaluate and adapt pedagogical strategies independently. Ultimately, generative AI tools will make teacher become an executor of AI's plans.

Proposed Solution Summary This project proposes an AI-mediated teacher decision support system based on the Cognitive Apprenticeship framework (Collins et al., 1986), rather than replacing teachers.

- **Input:** Lesson information and student learning data (e.g., Kahoot, Quizizz, Google Forms, CSV).
- **Analysis & Generation:** Our system will identify learning patterns, potential learning needs, and generate recommendations.
- **The 'Why' Effect:** Our system will explain the reasoning behind every recommendation, show how student data influenced the recommendation, how it relates to learning outcomes, and why a particular teaching strategy may be appropriate in the current situation.
- **Interaction & Output:** Teachers can accept, reject, modify, or annotate AI-generated suggestions, then they can build and export a final lesson plan.

By making expert reasoning visible, the system aims not only to reduce the effort required to interpret student data but also to help pre-service teachers develop data literacy, lesson planning skills, and professional judgement over time.

1.2 Review of Existing Systems

System 1: Generative AI Lesson Planning Tools

- **Problem Solved:** Generative AI lesson planning tools can help teachers create lesson plans, classroom activities, assessment tasks, and teaching resources quickly, thereby alleviating teachers’ problems of high pressure and tight timelines in preparing lessons. Representative frameworks include:
    - *ChatGPT:* An LLM that has parsed extensive web indexes and books (OpenAI, 2026). Teachers can directly send data to ChatGPT and receive suggestions.
    - *MagicSchool AI:* An AI platform developed specifically for teachers and school administrators (MagicSchool AI, 2026) where educators do not need to think about complicated prompts and only need to input parameters such as grade level and topic into form fields.
- **Strengths for Reference:** A major strength of these tools is their flexibility and adaptability. They can generate lesson plans, activities, assessment tasks, explanations, and feedback for different subjects, year levels, and learning contexts (Denny et al., 2024).
- **Limitations Addressed by Our Project:** Most generative AI lesson planning tools focus on producing outputs rather than explaining the outputs. It remains a black box to teachers. Teachers may receive a complete lesson plan without understanding how the recommendation relates to student data. This may limit the pedagogical development of pre-service and novice teachers. Our project addresses this limitation by treating teachers as active decision makers rather than passive users of AI-generated content. In order to overcome the limitations and preserve the advantages of Generative AI lesson planning tools, Our system will explain the reasoning why make the recommendations. Our System will also encourage teachers to review, compare, and modify suggestions before making final decisions. Making thinking visible is the principle of our system.

System 2: Learning Analytics Dashboards

- **Problem Solved:** Learning analytics dashboards are good tools for teachers to visualise and manage student learning data. Representative platforms include:
  - *Canvas Analytics:* Records students' comprehensive performance and learning behaviours in the whole semester or course process (Canvas LMS, 2026).
  - *Kahoot Reports:* A gamified interactive learning platform. After playing games in class or after class, it can generate reports to provide useful reference data for checking students' learning situation (Kahoot, 2026).
- **Strengths for Reference:** A major strength of these systems is their ability to present large amounts of learning data in a clear and accessible way (Canvas LMS, 2026). Data visualisation and summary reports can help teachers in many ways, such as identify students' overall grasp of knowledge.
- **Limitations Addressed by Our Project:** A main limitation of these systems is that they only show past data, but cannot actively support teachers to make next teaching decisions. Although they can help teachers in many ways, such as find students' learning problems, they seldom tell teachers what to do next. They also rarely explain why a teaching method is suitable. Our project aims to solve this problem and connect data analysis with real teaching. When finding a learning problem, the system will give some teaching suggestions, explain the reasons, and show how they relate to students' data. Teachers still make the final decisions, but the system can provide useful help in the whole teaching process.

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2. User Stories and Sprints

2.1 Scope Statement

The prototype will support input lesson context. We are going to deliver a web-based AI-mediated lesson planning mentor for pre-service teachers. It can also support uploading CSV or Excel student learning data. It will support data mapping and learning analytics. It can generate recommendations with expert reasoning explanations. Teachers can make decisions like accept, reject, modify, or annotate. It allows for the construction of an editable lesson plan. It keeps a revision history and supports Word or PDF export. The system won’t work as an automated lesson generator. It won’t use identifiable student data or integrate with live school systems.

Out of Scope

The following items are outside the scope of this project:

- **Live system integration:** The prototype won’t integrate with live UNSW, school, Department of Education, or learning management systems. It will be designed as a proof of concept using uploaded CSV or spreadsheet data.

- **Identifiable student data:** The system will only use synthetic or de-identified student learning data. It won’t ask for identifiable student information.

- **Automated lesson plan generation without teacher review:** The system won’t generate a final lesson plan without teacher decision points. Teachers have to review, accept, reject, modify, or add notes to AI-generated recommendations before using them in the lesson plan.

- **Replacing teacher judgement:** The system won’t make final pedagogical decisions for the teacher. The teacher is still responsible for understanding the data, evaluating AI suggestions, and creating the final lesson plan.

Client Approval Evidence

The project scope has been confirmed with the client, Sara Mashayekh, through email. After the first discussion with the client, the Product Owner sent an email to confirm the proposed scope, Demo 1 expectations, major features, and items that are out of scope.

The client replied that the proposed scope aligns well with her expectations. She also confirmed that the project should focus on student data analysis, AI-supported recommendations, teacher review and control, editable lesson plan construction, and export functionality.

So, our first demo will be a complete but simple functional prototype. It has these functions: input lesson background information, upload and map student data, do basic learning data analysis, get AI suggestions with clear reasons, let teachers check and make decisions, make lesson plans, and export files. The client said Demo 1 needs to show the whole work process from start to end.

Below is the screenshot of the client’s confirmation email, as proof that the client agrees with this plan.

Sara Mashayekh
收件人: Qianyi Zhang

答复 全部答复 转发
周三 2026/6/17 15:47

你已在 周三 2026/6/17 16:58 答复

此消息的语言为 英语 翻译至 中文 (简体) 始终不翻译 英语

Dear Qianyi and Team,
Thanks for your response.
Overall, the proposed scope aligns well with my expectations for the project and provides a strong foundation for your development planning. I am pleased to see the emphasis on student data analysis, AI supported recommendations, teacher review and control, editable lesson plan construction, and export functionality. The scope also appropriately excludes automated lesson planning without teacher review and maintains teacher decision making as a core component of the workflow.
For Group B, I would expect the system to include:

- visible reasoning or explanation for recommendations
- an expert or mentor reasoning panel where appropriate
- clear explanations of how student learning data informs recommendations
- worked examples or examples of effective planning decisions where feasible
- teacher accept/reject/modify/annotate workflows
- opportunities for users to compare their decisions with the system’s reasoning

The AI should function primarily as a mentor that makes planning expertise visible, rather than simply providing recommendations without explanation.
For Demo 1, your proposed direction is appropriate. I would encourage you to focus on demonstrating a complete end to end workflow that includes:

- lesson context input
- student data upload and mapping
- basic learning analytics
- AI supported recommendations
- visible reasoning behind recommendations
- teacher review and decision making
- lesson plan construction
- export functionality

One minor suggestion is that the wording “AI supported recommendations with expert reasoning explanations” captures the Group B focus particularly well and should remain a prominent part of the project description.
With those considerations in mind, I am happy for you to proceed with development planning based on the proposed scope.
I look forward to seeing your proposal and design ideas develop.

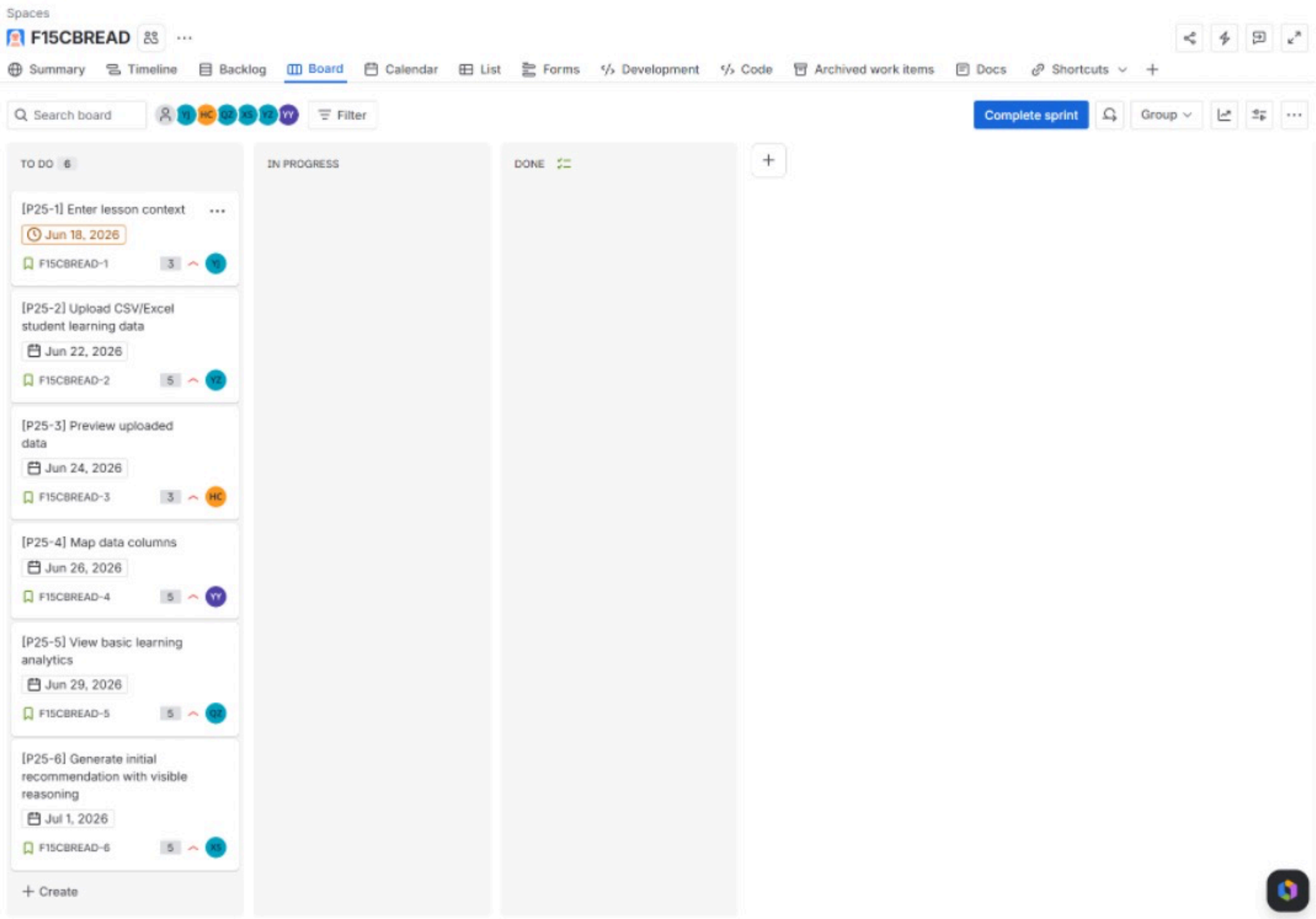
Best regards,
Sara

2.2 Sprint 1 User Stories

This section discusses the Scrum planning of MentorPlan AI. Sprint 1 is stuck in foundation and core enabling functionality, including lesson context input, CSV/Excel upload, data preview, column mapping, basic learning analytics and an initial recommendation explanation. More advanced features like human-in-the-loop decision review, AI assisted lesson plan generation, revision history, export, and the educator/researcher dashboard are in the product backlog for future sprint planning.

Total estimated product scope is 78 story points. Sprint 1 has 26 story points, which is approximately one-third of the total estimated work. We make this allocation to avoid both overcommitment and undercommitment and to ensure the team has a functional end-to-end foundation to show by the first progressive demo.

Jira Evidence:



Jira Key	User Story	Priority	Story Points
P25-1	As a pre-service teacher, I want to enter lesson context, so that recommendations are aligned with my teaching goals.	High	3
P25-2	As a pre-service teacher, I want to upload CSV/Excel student learning data, so that the system can analyse class learning patterns.	High	5
P25-3	As a pre-service teacher, I want to preview uploaded data, so that I can verify the dataset before analysis.	High	3
P25-4	As a pre-service teacher, I want to map data columns, so that the system can interpret different dataset formats.	High	5
P25-5	As a pre-service teacher, I want to view basic learning analytics, so that I can understand class strengths and areas needing support.	High	5
P25-6	As a pre-service teacher, I want early recommendations to include visible reasoning, so that I can understand why a suggestion is made.	High	5

Sprint 1 Acceptance Criteria

Jira Key	Acceptance Criteria
P25-1	Given I am on the lesson context page, when I enter the year level, subject area, topic, learning outcomes, lesson goals, learner needs, and constraints, then the system saves the lesson context and makes it available for later analytics and recommendation generation. If required fields are missing, the system displays a clear validation message.
P25-2	Given I am on the data upload page, when I upload a valid CSV or Excel file using synthetic or de-identified student learning data, then the system accepts the file and stores the dataset metadata. If the file type is invalid or the file is empty, the system displays an appropriate error message.
P25-3	Given a dataset has been uploaded successfully, when I open the data preview page, then the system displays the first rows and column names of the dataset. The preview must allow the teacher to confirm whether the uploaded data is correct before proceeding to mapping and analysis.
P25-4	Given the uploaded dataset contains columns such as student ID, question or item number, score, topic, skill area, learning outcome, or misconception category, when I map these columns to the required system fields, then the system validates the mapping and stores it. If required mappings are missing, the system prevents analysis and explains what must be fixed.
P25-5	Given the dataset has been uploaded and mapped successfully, when I run the basic analytics module, then the system displays class-level learning patterns including class average, low-scoring items, topic performance, common areas of difficulty, and possible extension opportunities. The analytics language must avoid deterministic labels such as “weak students”.
P25-6	Given basic analytics results are available, when the system generates an initial teaching recommendation, then the recommendation includes a short explanation of the supporting data evidence and expert lesson planning reasoning. The teacher should be able to understand why the suggestion was made, even if the full accept/reject/modify workflow is implemented in a later sprint.

2.3 Product Backlog

The product backlog below contains user stories not yet committed to Sprint 1. These stories are not yet assigned to a sprint as per Scrum progressive planning principles. Based on Sprint 1 outcomes, team velocity, client feedback and technical dependencies, they will be prioritised and selected for Sprint 2 or Sprint 3.

Jira Evidence:

Backlog (12 work items) 52 0 0 Create sprint

F15CBREAD-7	[P25-7] View risk and opportunity summary	TO DO	5
F15CBREAD-8	[P25-8] Generate student grouping suggestions with rationale	TO DO	5
F15CBREAD-9	[P25-9] Generate differentiation recommendations with expert ...	TO DO	5
F15CBREAD-10	[P25-10] Show outcome-activity-assessment alignment	TO DO	3
F15CBREAD-11	[P25-11] Accept, reject, modify, or annotate AI recommendati...	TO DO	5
F15CBREAD-12	[P25-12] Provide teacher reflection prompts	TO DO	3
F15CBREAD-13	[P25-13] Generate AI-supported draft lesson plan from appro...	TO DO	8
F15CBREAD-14	[P25-14] Edit generated lesson plan	TO DO	3
F15CBREAD-15	[P25-15] Export final lesson plan	TO DO	3
F15CBREAD-16	[P25-16] View revision history	TO DO	3
F15CBREAD-17	[P25-17] View anonymised educator or researcher dashboard	TO DO	5
F15CBREAD-18	[P25-18] Document prompt design, decision logic, testing, an...	TO DO	4

Jira Key	User Story	Priority	Story Points	Status	Acceptance Criteria Summary
P25-7	As a pre-service teacher, I want to view a risk and opportunity summary, so that I can identify possible learning needs before planning the lesson.	High	5	Backlog	The system summarises class-wide risks, strengths, possible support needs, and extension opportunities using careful, non-deterministic language.
P25-8	As a pre-service teacher, I want student grouping suggestions with rationale, so that I can form groups based on learning needs, mixed ability, or extension opportunities.	High	5	Backlog	The system suggests possible student groups and explains the data evidence and pedagogical reasoning behind each grouping. Teachers can later edit or reject groups.
P25-9	As a pre-service teacher, I want differentiation recommendations with expert reasoning, so that I can adapt activities for different learner needs.	High	5	Backlog	The system recommends strategies such as reteaching, scaffolding, targeted practice, peer support, and extension, with reasoning linked to analytics and learning outcomes.
P25-10	As a pre-service teacher, I want to see how each recommendation aligns with learning outcomes, activities, and assessment, so that I can understand the expert planning logic behind the recommendation.	High	3	Backlog	Each recommendation includes learning outcome alignment, activity rationale, and assessment alignment.

Jira Key	User Story	Priority	Story Points	Status	Acceptance Criteria Summary
P25-11	As a pre-service teacher, I want to accept, reject, modify, or annotate AI-generated recommendations, so that I remain responsible for pedagogical decisions.	High	5	Backlog	Recommendation cards provide accept, reject, modify, and annotation actions. Teacher decisions are saved for later lesson plan generation and revision history.
P25-12	As a pre-service teacher, I want teacher reflection prompts, so that I can justify key pedagogical decisions.	Medium	3	Backlog	The system prompts teachers to explain why they accepted, rejected, or changed suggested groupings, differentiation strategies, or lesson priorities.
P25-13	As a pre-service teacher, I want an AI-supported draft lesson plan based on approved decisions, so that I can construct a data-informed lesson plan efficiently.	High	8	Backlog	The system generates a draft lesson plan using teacher context, analytics summary, approved recommendations, and teacher decisions. The draft must not be generated directly from raw CSV data alone.
P25-14	As a pre-service teacher, I want to edit the generated lesson plan, so that the final plan reflects my professional judgement.	High	3	Backlog	The lesson plan editor allows teachers to revise lesson objectives, activities, differentiation notes, assessment tasks, and teacher notes before export.
P25-15	As a pre-service teacher, I want to export the final lesson plan, so that I can use it outside the system.	Medium	3	Backlog	The system exports the final lesson plan in a usable format such as Word, PDF, or structured text.
P25-16	As a pre-service teacher, I want to view revision history, so that I can see how the AI suggestions changed after teacher review.	Medium	3	Backlog	The system records the original AI recommendation, teacher decision, modified content, and final lesson plan component.
P25-17	As an educator or researcher, I want to view anonymised teacher interaction patterns, so that I can understand how teachers accept, reject, modify,	Medium	5	Backlog	The dashboard shows anonymised counts and summaries of teacher interactions with AI recommendations. No

Jira Key	User Story	Priority	Story Points	Status	Acceptance Criteria Summary
	or annotate AI recommendations.				identifiable student information is displayed.
P25-18	As a project stakeholder, I want prompt design, decision logic, testing, and deployment documentation, so that the prototype can be demonstrated and handed over transparently.	Medium	4	Backlog	The team documents prompt structure, analytics logic, recommendation rules, assumptions, limitations, testing outcomes, and deployment or demonstration instructions.

2.4 Sprint Timeline and Milestones

The project will be split into 3 main sprints. Sprint 1 is about the basic functionality. We expect that Sprint 2 will have the largest feature workload which will include recommendation reasoning, human-in-the-loop review and AI lesson plan generation. The team will also prepare for final demo, final report, software quality submission, documentation and project handover, so there is less development workload in Sprint 3.

The assessment guidance recommends allocating roughly one third of the total estimated story points to Sprint 1, prioritising foundation functionality in Sprint 1, major feature integration in Sprint 2, and finalisation, testing, documentation and handover in Sprint 3. This informs the sprint allocation.

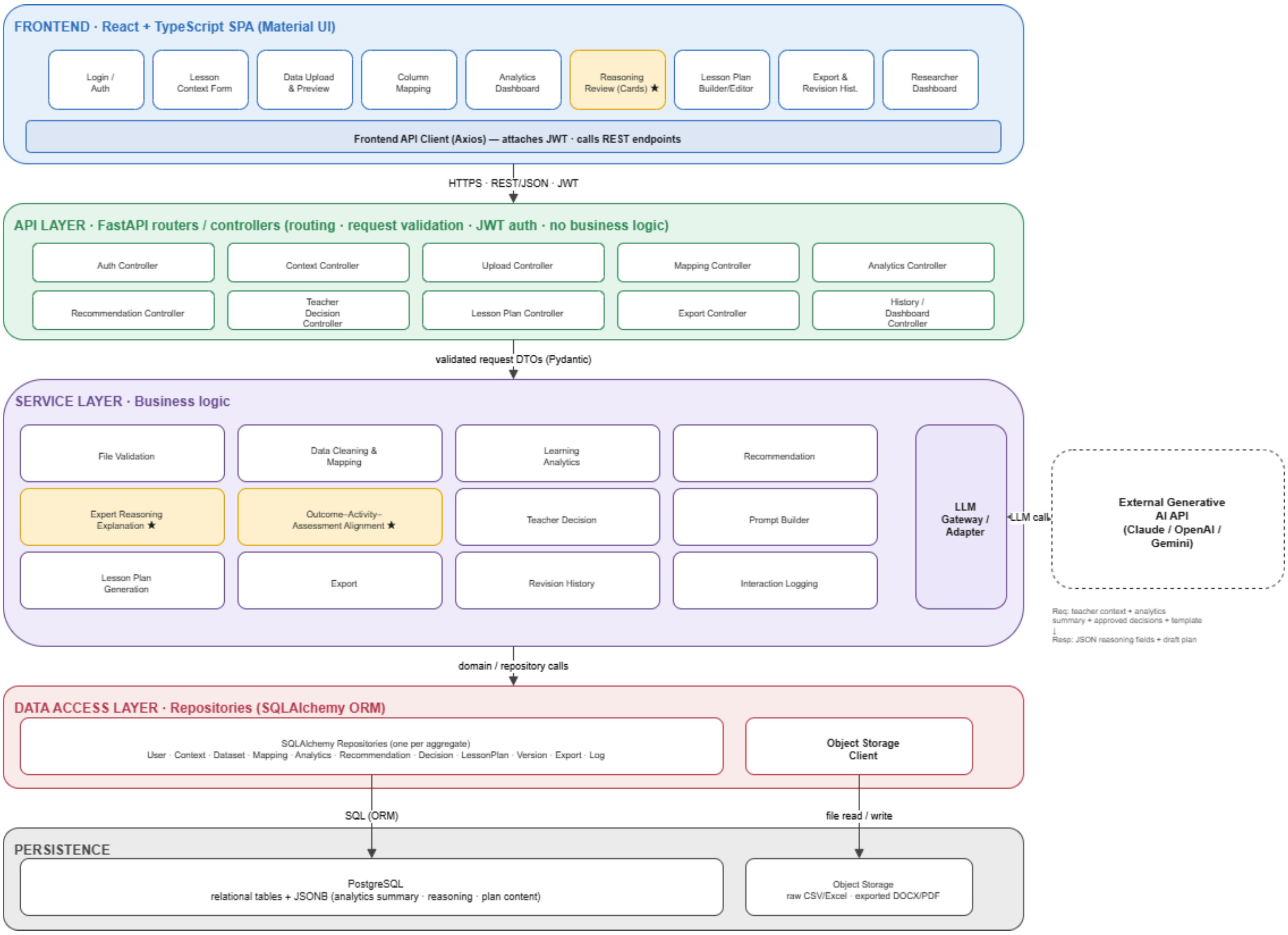
Sprint	Dates	Sprint Goal	Key Milestones / Deliverables
Sprint 1	Week 3 Wednesday after proposal submission – Week 5 Lab Time	Build the core foundation of the teacher workflow.	Lesson context form completed; CSV/Excel upload implemented; data preview implemented; column mapping prototype completed; basic analytics dashboard implemented; initial recommendation with visible expert reasoning generated; Sprint 1 review and retrospective prepared.
Sprint 2	Week 5 Lab Time – Week 8 Lab Time	Implement major recommendation, reasoning, and human-in-the-loop functionality.	Risk and opportunity summary implemented; grouping recommendation with rationale implemented; differentiation recommendation with expert reasoning implemented; outcome-activity-assessment alignment explanation implemented; accept/reject/modify/annotate workflow implemented; teacher reflection prompts implemented; AI-supported lesson plan generation prototype completed; Sprint 2 progressive demo and retrospective prepared.

Sprint	Dates	Sprint Goal	Key Milestones / Deliverables
Sprint 3	Week 8 Lab Time – Week 10 Lab Time	Finalise export, revision history, dashboard, testing, deployment, and documentation.	Lesson plan editor completed; export function implemented; revision history implemented; educator/researcher dashboard implemented; prompt and decision logic documentation completed; synthetic dataset and demonstration workflow prepared; system testing and UI polish completed; final demo, final report, software quality submission, and handover prepared.

3. Technical Design

3.1 System Architecture and Backend/Data Design

3.1.1 System Architecture Diagram - Overview



Dashed boxes represent third-party services.

The System Architecture Diagram presents the layered architecture of MentorPlan AI. The frontend handles teacher-facing workflows, while the FastAPI controller layer only performs routing, JWT authentication, and request validation. Business logic is placed in the service layer, including data cleaning, learning analytics, recommendation generation, expert reasoning, teacher decisions, prompt building, lesson plan generation, revision history, export, and interaction logging.

3.1.2 Component Description Table

Layer	Component	Responsibility	Key Data / Interaction
Frontend	React + TypeScript Web App	Login, context input, upload, mapping, analytics, reasoning display, editor, export, history, dashboard	Axios + JWT calls backend APIs
API	FastAPI Controllers	Authentication, context, upload, mapping, analytics, recommendations, decisions, plans, export	Routing, validation, and authentication only
Service	Business Modules	File checks, cleaning, analytics, recommendations, decisions, prompts, plans, exports, versions, logs	Runs the main workflow
Core	Reasoning + Alignment	Evidence, expert logic, outcome/activity/assessment fit, differentiation	Shows AI thinking to teachers
AI Boundary	AI Adapter + LLM APIs	Connects to LLMs and validates structured output	No raw student files are sent
Data Access	Repositories + Storage Tools	One repository per domain object	Only layer that reads and writes data
Persistence	PostgreSQL + Object Storage	Database for forms and JSON; file storage for uploads and exports	Large files stay outside the database

3.1.3 API / Data Flow Explanation

Step	Example API	Data Flow
1. Context + Upload	Create context; upload dataset	Lesson context is saved to the database; raw CSV/Excel files are saved to file storage
2. Mapping + Cleaning	Map columns; validate data	Dataset columns are mapped to standard fields; cleaned student responses are saved
3. Analytics	Get dataset analytics	Class patterns, low-scoring areas, and common mistakes are saved as JSON
4. Recommendations	Get lesson suggestions	Teaching strategies are generated with evidence and alignment notes
5. Teacher Decisions	Submit decision records	Accept, reject, edit, and comment actions are saved
6. Plan + Export	Generate, update, and export lesson plan	AI uses summaries and approved actions; lesson plans and exported files are saved
7. Research Dashboard	Get interaction statistics	Anonymous teacher actions are summarised for researchers

3.1.4 Initial Database Entity List

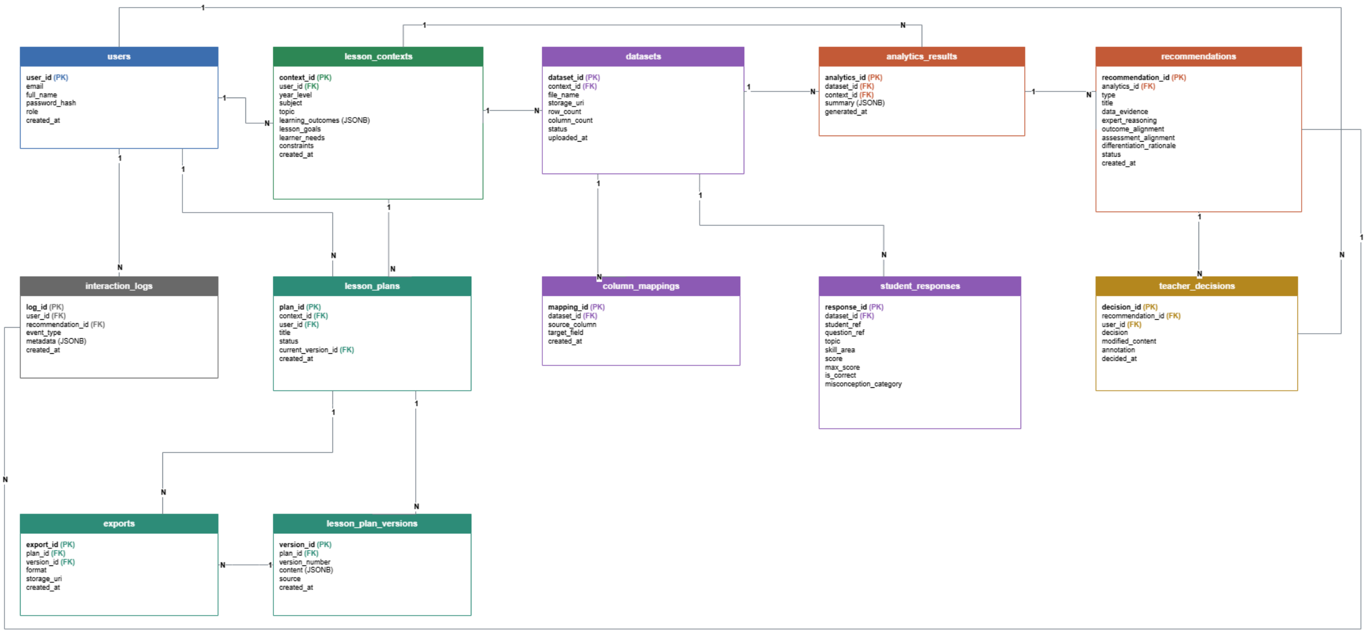


Table	Key Fields	Purpose
`users`	`user_id`, `email`, `role`	Teacher and researcher accounts
`lesson_contexts`	`context_id`, `user_id`, `year`, `subject`, `topic`, `outcomes JSON`	Basic lesson context
`datasets`	`dataset_id`, `context_id`, `file name`, `storage URI`, `rows`, `status`	Uploaded file metadata
`column_mappings`	`mapping_id`, `dataset_id`, `source column`, `target field`	Column-to-field mapping
`student_responses`	`response_id`, `dataset_id`, `student ref`, `topic`, `score`, `max`, `correct`	Anonymised response data
`analytics_results`	`analytics_id`, `dataset/context id`, `summary JSON`	Class analytics
`recommendations`	`recommendation_id`, `analytics_id`, `evidence`, `reasoning`, `alignment`, `differentiation`	AI suggestions and reasoning
`teacher_decisions`	`decision_id`, `recommendation/user id`, `decision`, `edits`, `notes`	Teacher review actions
`lesson_plans`	`plan_id`, `context/user id`, `current version`, `status`	Current lesson plan
`lesson_plan_versions`	`version_id`, `plan_id`, `version`, `content JSON`, `source`	Plan revision history
`exports`	`export_id`, `plan/version id`, `format`, `storage URI`	Exported Word/PDF records

Table	Key Fields	Purpose
`interaction_logs`	`log_id`, `user/recommendation id`, `event`, `metadata JSON`	Anonymous research logs

3.1.5 Privacy and AI Boundary

Privacy is treated as a strict design requirement.

Raw student files remain in object storage and are not sent directly to the AI model. The AI only receives summaries, approved teacher actions, and templates. Teachers review AI suggestions before those suggestions are used in final lesson plans.

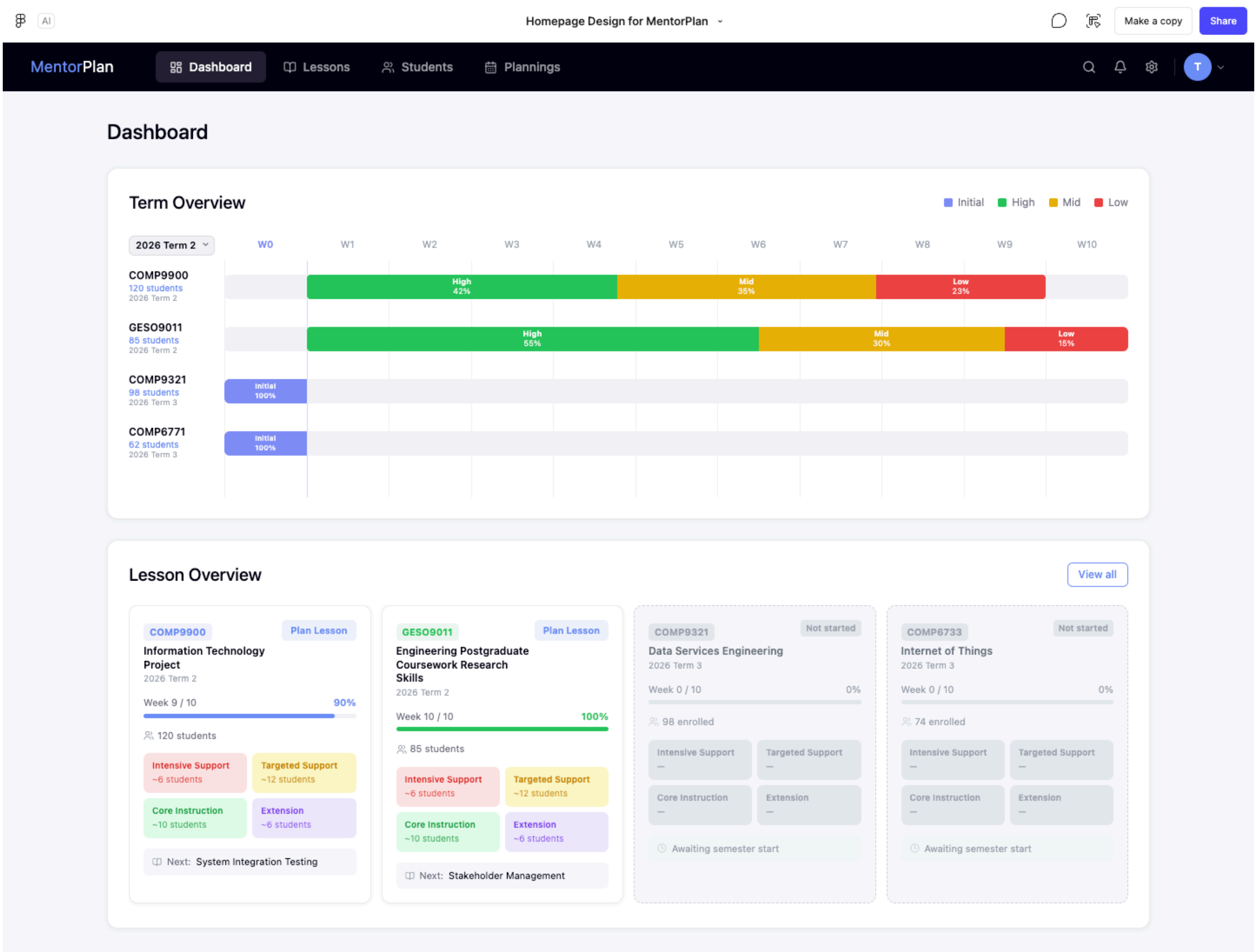
This design reduces privacy risk, supports teacher agency, and ensures that AI remains a decision-support tool rather than a replacement for teacher judgement.

3.2 Interface Storyboarding

To illustrate the proposed user interactions and frontend layout of MentorPlan AI, we developed a set of high-fidelity interface storyboards. These storyboards demonstrate the major system functions, including course overview, lesson management, student data management, AI-supported data analysis, recommendation generation, and lesson plan export.

The interface follows a dashboard-based web application structure. A persistent top navigation bar allows teachers to move between Dashboard, Lessons, Students, and AI Planner. The AI Planner uses a step-by-step workflow: **Data & Analysis** → **Recommendations** → **Review & Export**. This design helps teachers progressively move from reviewing student data to understanding AI reasoning and finally exporting a teacher-approved lesson plan.

3.2.1 Dashboard Page



Covered user stories: P25-5, P25-7, P25-8, P25-17

The Dashboard page provides teachers with a high-level overview of their teaching term, courses, student performance distribution, and lesson planning progress. The page uses a clean card-based layout with a dark top navigation bar and a light background to separate navigation from working content.

The upper section, Term Overview, visualises multiple courses across weekly timelines. Each course row shows the current stage of the term and the distribution of student learning levels, such as Initial, High, Mid, and Low. This allows teachers to quickly identify which courses are active, which ones have not started, and which groups may require further support.

The lower section, Lesson Overview, displays course cards. Each card summarises course name, term, week progress, completion percentage, number of students, and support group estimates. For active courses, the card shows support categories such as Intensive Support, Targeted Support, Core Instruction, and Extension. For future courses, the card is visually disabled and marked as not started.

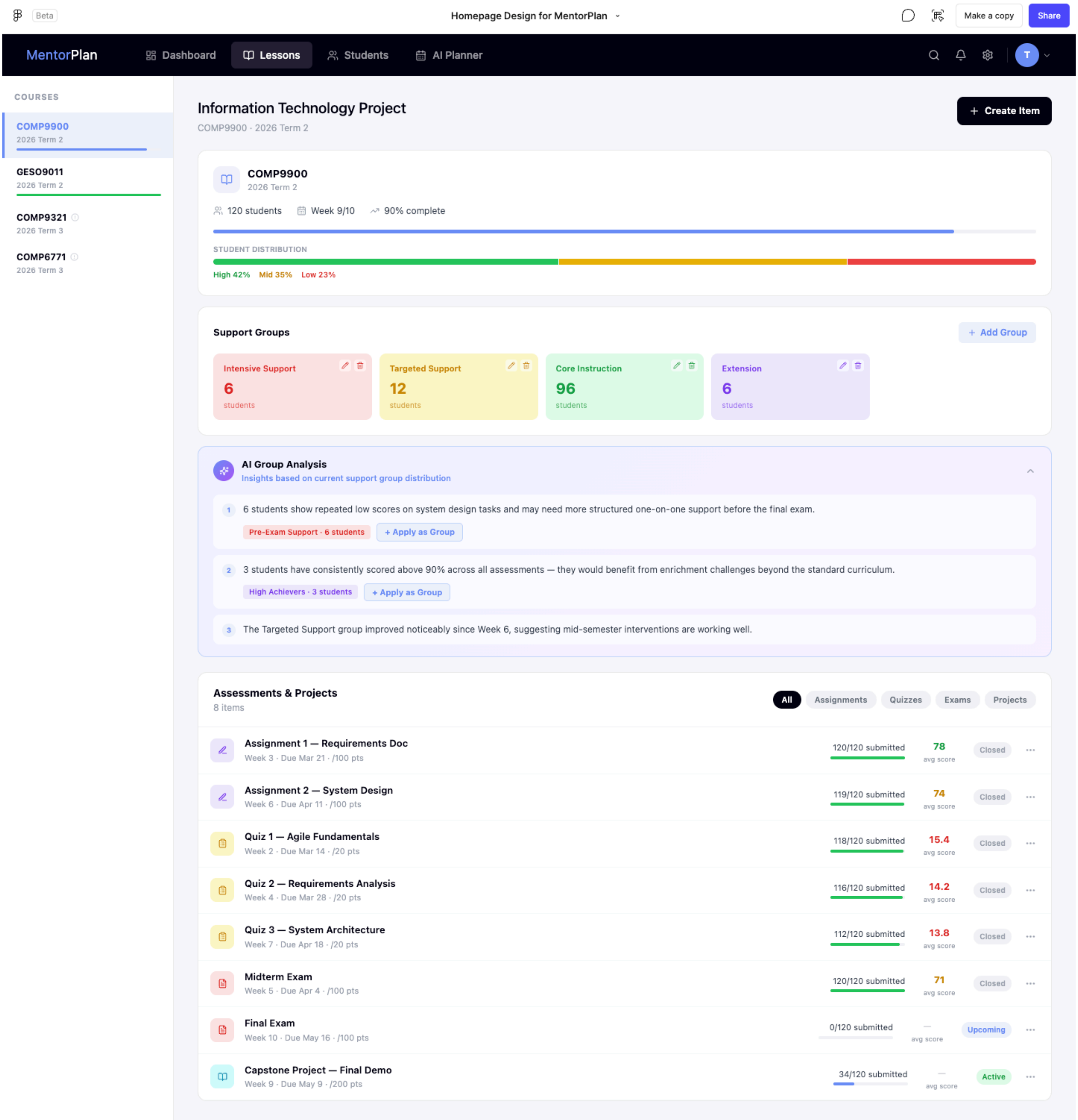
This storyboard supports the system objective of helping teachers interpret class-level learning patterns quickly. It also provides an entry point into more detailed lesson planning and support group analysis.

Main interactions:

- The teacher can switch between courses using the dashboard cards.
- The teacher can view the current term progress and course completion status.
- The teacher can identify which student support groups may need attention.

- The teacher can click into a course or lesson card to continue planning.

3.2.2 Lessons Page



Covered user stories: P25-5, P25-7, P25-8, P25-9, P25-10

The Lessons page provides a course-level view for a selected subject. A left sidebar lists available courses, while the main content area shows detailed information about the selected course. This layout allows teachers to switch between courses without leaving the lesson management workflow.

At the top of the main panel, the selected course card displays course code, term, number of students, current week, completion percentage, and overall student distribution. This gives teachers a quick understanding of the course context before making planning decisions.

The Support Groups section summarises student groupings into categories such as Intensive Support, Targeted Support, Core Instruction, and Extension. Each support group card shows the number of students in that group and provides editing or management actions. This supports teacher control over AI-assisted grouping rather than forcing teachers to accept system-generated categories.

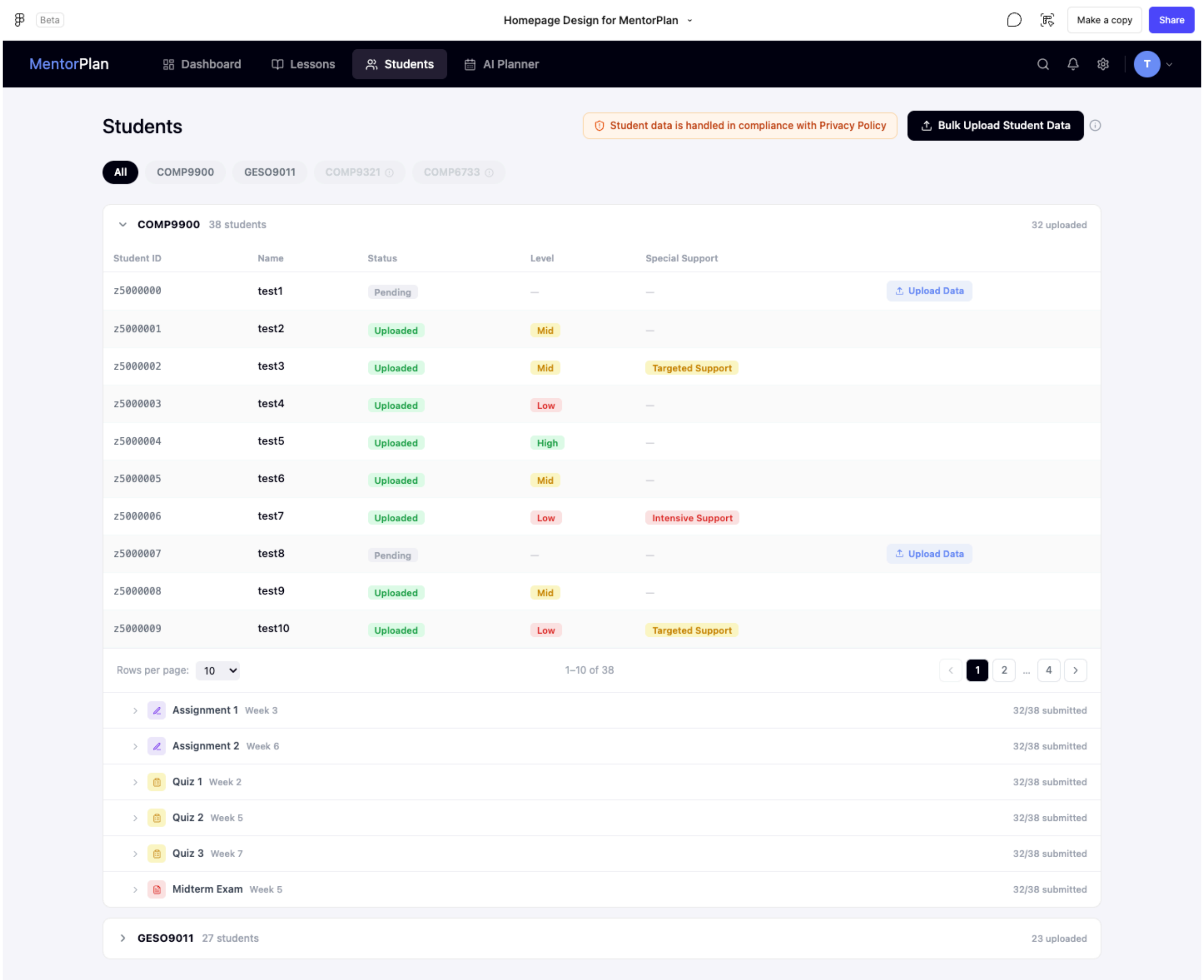
The AI Group Analysis section presents reasoning based on the current support group distribution. It explains patterns such as repeated low scores, high-achieving students requiring enrichment, or noticeable improvement after previous interventions. These explanations help make the AI reasoning process visible and interpretable.

The Assessments & Projects section lists assessment items, quizzes, exams, and projects. Each row shows submission progress, average score, and status. This helps teachers connect student data sources with lesson planning decisions.

Main interactions:

- The teacher can select different courses from the sidebar.
- The teacher can review support group distribution.
- The teacher can add, edit, or apply support groups.
- The teacher can inspect AI-generated group analysis.
- The teacher can view assessments and projects as evidence sources for planning.

3.2.3 Students Page



Covered user stories: P25-2, P25-3, P25-4, P25-5

The Students page focuses on student data management and upload status. It supports the data preparation stage before analytics and AI planning. The page includes course filters, privacy notice, student table, upload actions, and assessment submission summaries.

At the top of the page, course filter tabs allow teachers to view all students or only students from a specific course. A privacy notice reminds users that student data is handled according to privacy policy requirements. This reinforces the project's design principle that only synthetic or de-identified student data should be used.

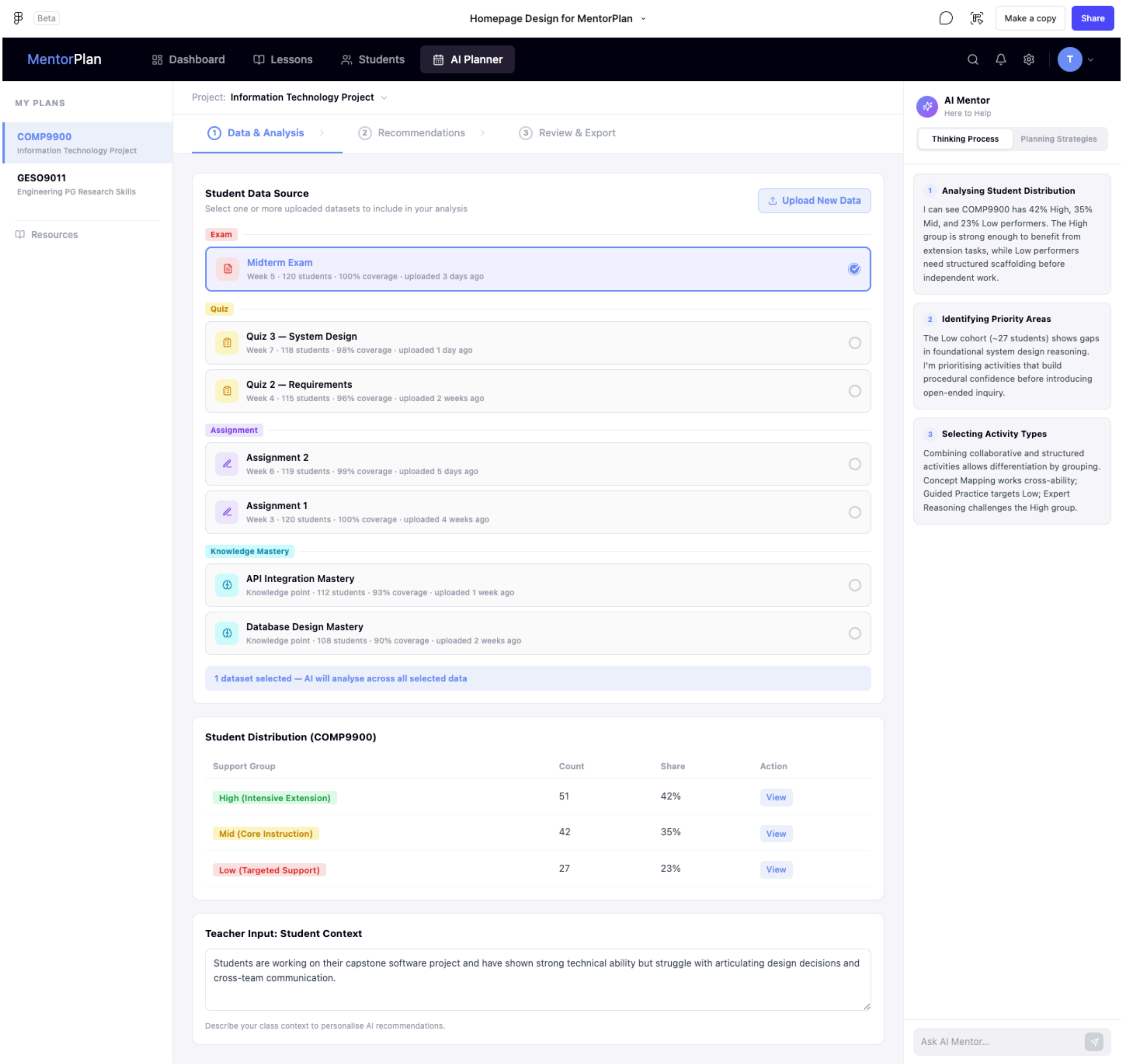
The main student table shows student ID, name, upload status, performance level, and special support category. Students with uploaded data are marked as Uploaded, while those without submitted data are marked as Pending and provide an Upload Data action. This helps teachers identify missing datasets before running analytics.

The lower part of the page lists assessment items such as assignments, quizzes, and exams. Each item includes submission progress, such as how many students have submitted data. This supports the data preview and verification workflow because teachers can check whether enough student data is available before proceeding to analysis.

Main interactions:

- The teacher can filter students by course.
- The teacher can bulk upload student data.
- The teacher can upload data for individual students.
- The teacher can review upload status and identify missing submissions.
- The teacher can expand assessment rows to inspect submission coverage.

3.2.4 AI Planner: Data & Analysis Page



Covered user stories: P25-1, P25-2, P25-3, P25-4, P25-5, P25-6

The AI Planner begins with the Data & Analysis step. This screen allows teachers to choose the student data sources that will be used for AI-supported lesson planning. The left sidebar lists the teacher's active plans, while the top stepper shows the three-stage workflow: Data & Analysis, Recommendations, and Review & Export.

The Student Data Source section lists available datasets, including exams, quizzes, assignments, and knowledge mastery checkpoints. Each dataset card shows the data type, title, week, number of students, coverage percentage, and upload time. Teachers can select one or more datasets for analysis. This design gives teachers control over which evidence is used by the system.

The Student Distribution section shows a summary of student performance groups for the selected dataset. In the example, students are divided into High, Mid, and Low groups with counts and percentage shares. Teachers can also view the details of each group.

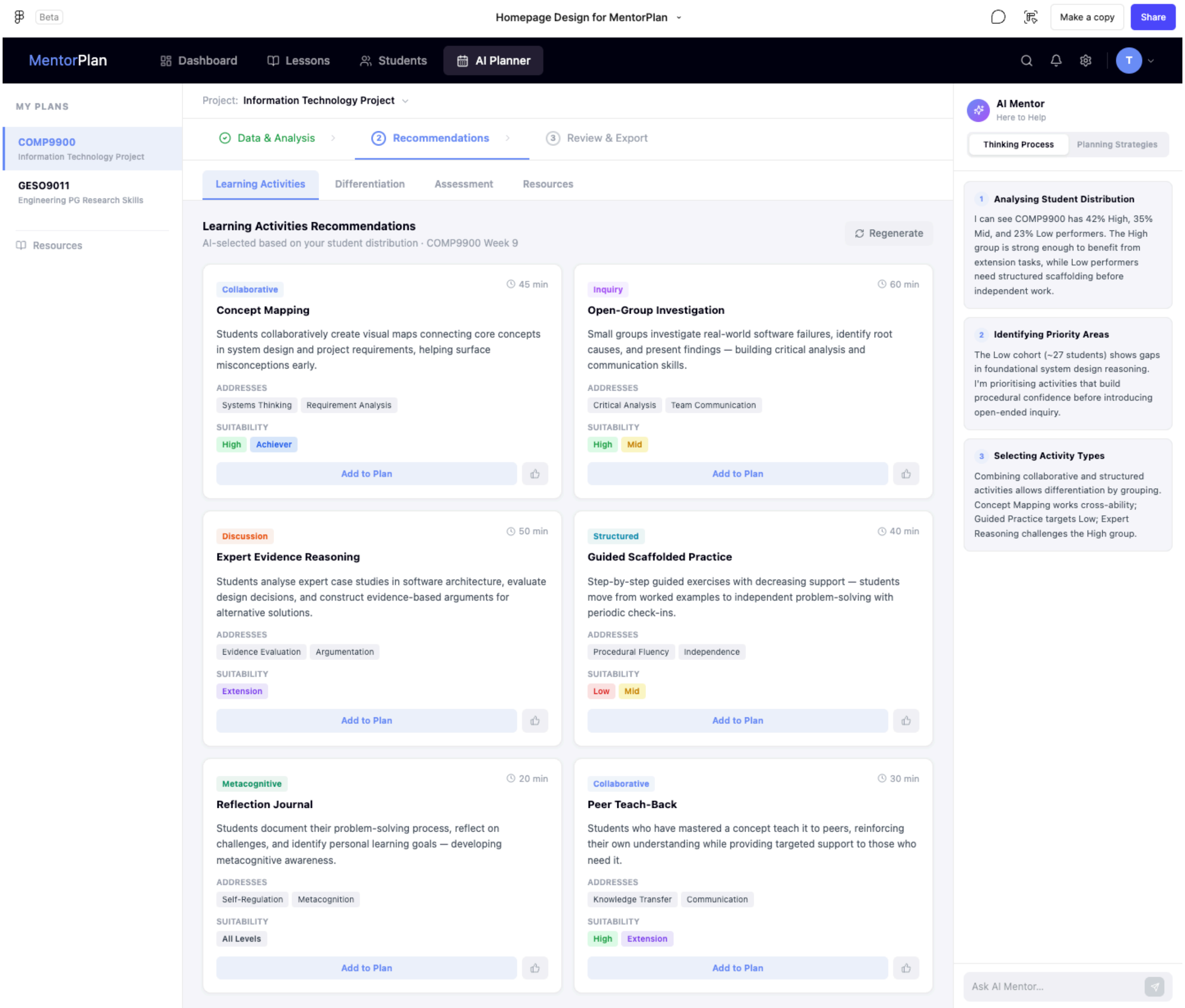
The Teacher Input: Student Context section allows teachers to add contextual information that cannot be fully captured by numerical assessment data, such as class background, communication issues, project progress, or learning constraints. This supports more personalised and pedagogically appropriate AI recommendations.

A right-side AI Mentor panel explains the thinking process. It summarises how the AI interprets student distribution, identifies priority areas, and selects suitable activity types. This sidebar supports the system's "why" effect by making the reasoning process visible rather than only showing final recommendations.

Main interactions:

- The teacher selects one or more uploaded datasets.
- The teacher reviews data coverage and student distribution.
- The teacher adds student context to personalise recommendations.
- The AI Mentor panel displays visible reasoning based on selected data.
- The teacher proceeds from data analysis to recommendation generation.

3.2.5 AI Planner: Recommendations Page



Covered user stories: P25-6, P25-8, P25-9, P25-10, P25-11, P25-12

The Recommendations page presents AI-generated lesson activity suggestions based on the selected student data and teacher-provided context. The page uses recommendation cards to display multiple possible teaching strategies instead of producing one final lesson plan immediately. This keeps the teacher in control of pedagogical decision-making.

The top stepper shows that the teacher has completed Data & Analysis and is now in the Recommendations stage. Recommendation categories include Learning Activities, Differentiation, Assessment, and Resources. This tab structure allows recommendations to be organised according to different lesson planning needs.

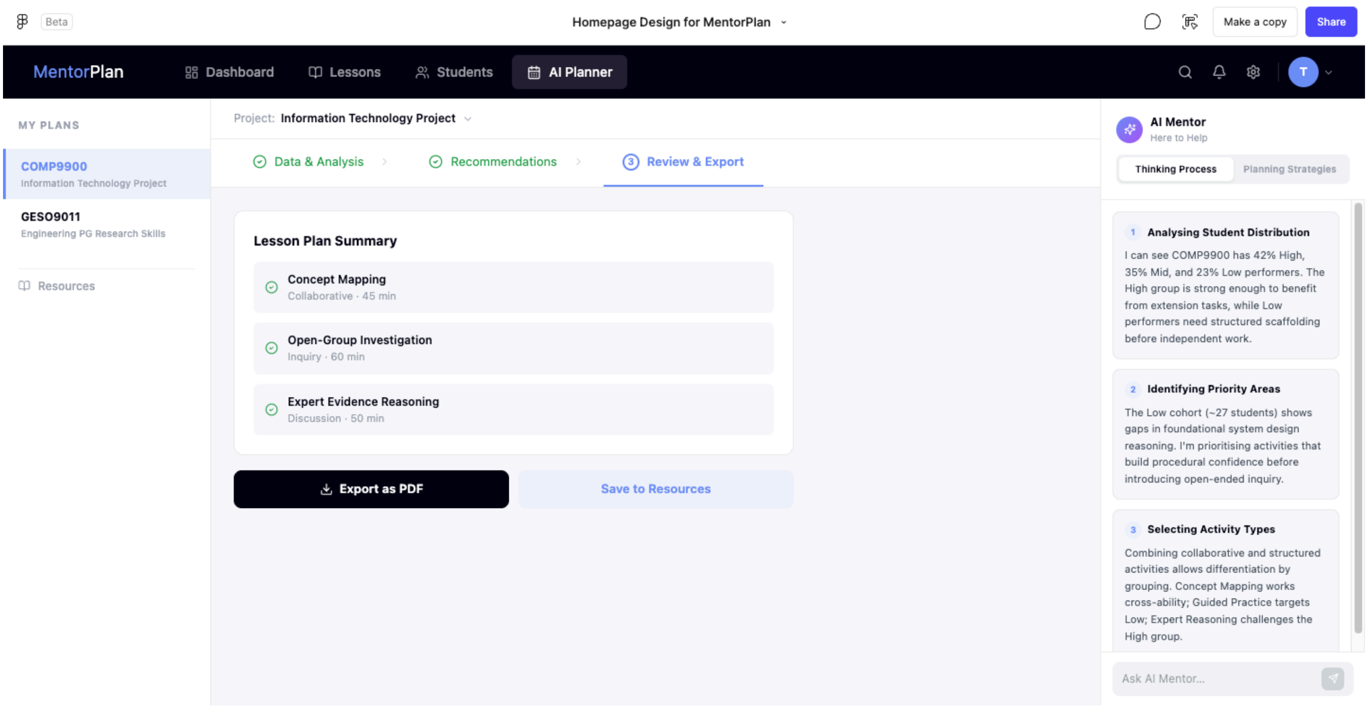
Each recommendation card contains an activity title, activity type, estimated duration, short description, addressed skills, suitability tags, and an Add to Plan button. Examples include Concept Mapping, Open-Group Investigation, Expert Evidence Reasoning, Guided Scaffolded Practice, Reflection Journal, and Peer Teach-Back. Suitability tags such as High, Mid, Low, Extension, or All Levels show which student groups the activity is designed for.

The AI Mentor panel remains visible on the right side of the page. It explains the reasoning process behind the recommendations, including how student distribution was analysed, how priority areas were identified, and why particular activity types were selected. This design directly addresses the limitation of black-box AI tools by making the recommendation logic visible.

Main interactions:

- The teacher switches between recommendation categories.
- The teacher reads the evidence and rationale behind each recommendation.
- The teacher adds selected recommendations to the lesson plan.
- The teacher can regenerate recommendations when needed.
- The teacher can use the AI Mentor panel to understand the reasoning behind the suggestions.

3.2.6 AI Planner: Review & Export Page



Covered user stories: P25-11, P25-13, P25-14, P25-15, P25-16

The Review & Export page is the final stage of the AI Planner workflow. It allows teachers to review selected lesson plan components before exporting or saving the final plan. This ensures that the output is based on teacher-approved decisions rather than generated directly from raw student data.

The main panel shows a Lesson Plan Summary containing selected activities such as Concept Mapping, Open-Group Investigation, and Expert Evidence Reasoning. Each item includes the activity type and estimated duration. The selected activities are shown in a concise summary format so that teachers can confirm whether the final plan matches their instructional goals.

Below the summary, teachers can choose to Export as PDF or Save to Resources. Exporting supports use outside the system, while saving to resources allows the plan to be reused or revised later. This supports both immediate classroom use and longer-term lesson planning workflows.

The AI Mentor panel remains visible on the right side, showing the reasoning trail that led to the selected plan. This helps teachers review whether the final lesson plan still aligns with the original data analysis and recommendation logic.

Main interactions:

- The teacher reviews selected lesson activities.
- The teacher confirms that the lesson plan reflects their decisions.
- The teacher exports the final plan as a PDF.
- The teacher saves the plan to resources for future use.
- The system preserves the planning and reasoning trail for revision history.

3.2.7 Overall Navigation Flow

The storyboarded workflow can be summarised as follows:

1. The teacher starts from the **Dashboard** to review course progress and support group distribution.
2. The teacher opens the **Lessons** page to inspect course-level assessments, support groups, and AI group analysis.
3. The teacher uses the **Students** page to upload or verify student data.
4. The teacher enters the **AI Planner** and selects relevant datasets in **Data & Analysis**.
5. The system generates interpretable recommendations in the **Recommendations** stage.
6. The teacher reviews selected items and exports the final plan in **Review & Export**.

This flow supports the project goal of building an AI-mediated teacher decision support tool. The system does not replace teacher judgement. Instead, it helps teachers interpret student data, understand the reasoning behind recommendations, make informed pedagogical decisions, and export a final lesson plan based on approved actions.

3.3 Design Justifications

3.3.1 Decomposing the Problem into Manageable Subproblems

The overall problem is that pre-service teachers may struggle to interpret student learning data and transform it into evidence-based lesson planning decisions. To make this problem manageable, the project is divided into the following subproblems:

1. **Data collection and preparation**

   Teachers need a way to upload student learning data from sources such as CSV or Excel files. Since different datasets may use different column names or formats, the system must support data preview, column mapping, validation, and cleaning.

2. **Learning analytics generation**

   The system needs to identify class-level patterns, low-scoring areas, common mistakes, strengths, risks, and extension opportunities from the uploaded data.

3. **Visible AI reasoning**

   Instead of simply generating a lesson plan, the system must explain why a recommendation is made, what data evidence supports it, and how it connects to learning outcomes, activities, and assessment.

4. **Teacher decision-making workflow**

Teachers must remain active decision-makers. They need to be able to accept, reject, edit, or annotate AI-generated recommendations before these suggestions are used in a final lesson plan.

5. **Lesson plan construction and export**

   Approved teacher decisions should be assembled into an editable lesson plan. The final plan should be exportable for use outside the system.

6. **Privacy and traceability**

   Since student data is involved, the system must avoid sending raw student files to AI models. It should also keep records of teacher decisions, plan versions, exports, and anonymised interaction logs.

3.3.2 Layered Architecture

A layered architecture is chosen because it separates responsibilities clearly across the frontend, API layer, service layer, data access layer, and persistence layer.

An alternative design would be to build a simpler monolithic backend where controllers directly process data, call AI services, and write to the database. This would be faster to implement at the beginning, but it would make the system harder to test, extend, and maintain. It would also increase the risk of mixing business logic, data access, and privacy-sensitive operations in the same component.

The chosen layered design is more suitable for this project because:

- Controllers remain lightweight and only handle routing, validation, and authentication.
- Services contain the main workflow and business logic.
- Repositories isolate database and file storage operations.
- The AI adapter hides differences between LLM providers.
- Privacy rules can be enforced at the service and AI boundary layers.

This design supports future extension, such as adding new data formats, new recommendation types, or new AI model providers.

3.3.3 Human-in-the-Loop AI Design

The project deliberately avoids generating a complete final lesson plan directly from raw data. Instead, the system first presents analytics and recommendations, then requires teacher review before a lesson plan is generated.

An alternative approach would be a one-click AI lesson plan generator. This would be convenient, but it would reproduce the black-box limitation of existing generative AI tools. Teachers might receive a polished output without understanding why particular activities or strategies were recommended.

The chosen human-in-the-loop design is more appropriate because:

- Teachers can inspect the evidence behind recommendations.
- Teachers can accept, reject, modify, or annotate AI suggestions.
- The final lesson plan is based on approved decisions rather than raw AI output.
- The system supports teacher agency and professional judgement.
- The reasoning process helps pre-service teachers learn how data informs lesson planning.

This aligns with the project's goal of supporting cognitive apprenticeship by making expert reasoning visible.

3.3.4 AI Boundary and Privacy Protection

The system uses a strict AI boundary. Raw student files are kept in object storage and are not sent directly to the AI model. The AI only receives lesson context, analytics summaries, approved teacher decisions, and templates.

An alternative approach would be to send the full uploaded CSV or Excel file directly to an LLM for analysis. This could reduce backend development effort, but it creates privacy risks and makes the system overly dependent on the AI model's interpretation of raw data.

The chosen design is safer and more controlled because:

- Raw student data remains inside the system.
- The backend can clean, validate, and summarise data before AI interaction.
- AI prompts are based on structured summaries rather than uncontrolled raw files.
- Teacher-approved decisions are used before final plan generation.
- The system can provide clearer audit trails and revision history.

This design reduces privacy risk and supports responsible AI use in educational contexts.

3.3.5 Structured Recommendation and Alignment Design

Recommendations are designed to include evidence, reasoning, differentiation, and outcome-activity-assessment alignment. This means each recommendation should explain not only what the teacher could do, but also why it is appropriate.

An alternative design would be to show only activity suggestions, such as "use group work" or "provide scaffolding". However, this would not adequately support pre-service teachers because it would not explain how the suggestion relates to student data or learning outcomes.

The chosen design provides additional value because:

- Teachers can see which data patterns triggered each recommendation.
- Recommendations are linked to learning outcomes and classroom activities.
- Differentiation strategies can be matched to different support groups.
- Teachers can compare multiple options before deciding.
- The system provides novel functionality beyond standard learning analytics dashboards.

This helps bridge the gap between data visualisation and pedagogical action.

3.3.6 Object Storage for Files and PostgreSQL for Structured Data

The system separates file storage from structured data storage. PostgreSQL stores user accounts, lesson contexts, mappings, analytics results, recommendations, teacher decisions, lesson plans, versions, exports, and logs. Object storage stores larger files such as raw CSV/Excel uploads and exported Word/PDF files.

An alternative approach would be to store all files directly inside the relational database. This would simplify storage in the short term, but it would make the database larger and less efficient.

The chosen design is preferable because:

- Large files are kept outside the database.
- Structured records remain easy to query.
- File metadata can still be tracked in PostgreSQL.
- Exported documents can be stored and retrieved efficiently.
- The design is easier to scale if more datasets or exports are added later.

3.4 Functionality Mapping

The following table shows the project objectives, main functions, and related user stories.

Project Objective /	Related User Stories	Independent Justification
Build a web-based lesson planning workflow	P25-1, P25-11, P25-13, P25-14, P25-15	P25-1 supports lesson context input, P25-11 ensures teacher review before recommendations are used, P25-13 supports draft lesson plan generation from approved decisions, P25-14 supports teacher editing, and P25-15 supports export. Together, they cover the workflow from lesson context input to lesson plan export.
Support upload and mapping of student data	P25-2, P25-3, P25-4	P25-2 supports CSV/Excel upload, P25-3 supports data preview, and P25-4 supports column mapping. These stories ensure the system can process uploaded student datasets from different sources and formats.
Analyse student learning patterns	P25-5, P25-7	P25-5 provides basic analytics such as class average, low-scoring items, topic performance, and common areas of difficulty. P25-7 provides a risk and opportunity summary to identify strengths, support needs, and extension opportunities.
Generate AI-supported recommendations with visible reasoning	P25-6, P25-8, P25-9, P25-10	P25-6 supports initial recommendations with visible reasoning, P25-8 supports grouping rationale, P25-9 supports differentiation reasoning, and P25-10 explains outcome-activity-assessment alignment. These stories directly support the cognitive apprenticeship focus.
Maintain teacher control through human-in-the-loop review	P25-11, P25-12, P25-16	P25-11 allows teachers to accept, reject, modify, or annotate recommendations. P25-12 adds reflection prompts, and P25-16 records revision history. Together, they ensure the system supports teacher judgement rather than replacing it.
Generate and export an editable lesson plan	P25-13, P25-14, P25-15	P25-13 generates a draft lesson plan using teacher context, analytics summary, approved recommendations, and teacher decisions. P25-14 allows editing, and P25-15 enables export in Word, PDF, or structured text.
Support educator/researcher review of teacher-AI interaction patterns	P25-17	P25-17 provides anonymised counts and summaries of teacher interactions with AI recommendations, such as accepted, rejected, modified, and annotated patterns. This supports evaluation of how teachers use AI support.
Document prompt design, decision logic, testing, and deployment	P25-18	P25-18 requires documentation of prompt structure, analytics logic, recommendation rules, assumptions, limitations, testing outcomes, and deployment instructions. This supports transparency, maintainability, and handover.
Protect student privacy	P25-2, P25-3, P25-17, P25-18	P25-2 requires synthetic or de-identified student learning data, P25-3 allows preview before analysis, P25-17 ensures the researcher dashboard displays anonymised interaction patterns without identifiable student information, and P25-18 documents privacy assumptions and data handling limitations.
Make expert reasoning visible	P25-6, P25-8, P25-9, P25-10, P25-12	P25-6, P25-8, P25-9, and P25-10 add reasoning explanations for recommendation, grouping, differentiation, and alignment features. P25-12 encourages teachers to reflect on their own decisions, supporting cognitive apprenticeship.
Avoid automated lesson planning and preserve teacher judgement	P25-11, P25-13, P25-14, P25-16	P25-11 requires teacher review before recommendations are used, P25-13 generates drafts only from approved decisions, P25-14 allows teacher editing, and P25-16 records AI suggestions and teacher modifications. These stories prevent the system from replacing teacher judgement.

3.5 Design Justifications

The system is designed based on the theory of cognitive apprenticeship. This project aims to develop an artificial intelligence-assisted curriculum planning guidance tool. It can help teachers create curriculum plans and show the logic behind them.

For example, the system explains the data evidence, the connection with learning outcomes, the reason for activity design, the choice of assessment methods, and the logic of differentiated teaching. In this way, novice teachers can better connect student learning data, learning outcomes, learning activities, assessment strategies, and learner needs.

3.5.2 Subproblem Breakdown and Proposed Solutions

Subproblem	Proposed Solution	Design Justification	Possible Limitation and Mitigation
Pre-service teachers may not see expert lesson planning reasoning.	Use the Expert Reasoning Explanation Service to provide a reasoning explanation for each recommendation.	This directly supports the cognitive apprenticeship principle of making expert thinking visible. The system does not only tell users what to do; it explains why a recommendation is appropriate, helping pre-service teachers understand how	LLM explanations may be generic. To mitigate this, the Prompt Builder Service will require each explanation to reference student data evidence, learning outcomes, and lesson goals.
Raw CSV/Excel student data is difficult to translate into pedagogical decisions.	Use the Learning Analytics Service to analyse class average, topic performance, low-scoring items, common misconceptions, possible support needs, and	Teachers need data-informed insights rather than raw score tables. The system first converts student data into understandable learning patterns before generating teaching recommendations.	Simple rules may miss complex learning patterns. The MVP prioritises transparent rules, while clustering or more advanced learning analytics can be added in future work.
AI recommendations may appear unsupported or arbitrary.	Use the Recommendation Service to generate recommendations with data evidence, expert reasoning, learning outcome alignment, and	This makes the recommendation process more transparent and distinguishes the system from generic AI lesson planners. Users can see how the system moves from data patterns to teaching suggestions.	Even with explanations, final adoption still requires teacher judgement. All outputs will be presented as suggestions rather than final decisions.
Learning activities and assessments may not clearly align with learning outcomes.	Use the Outcome-Activity-Assessment Alignment Service to explain how each recommendation connects learning outcomes, activities, and	This helps novice teachers understand how experienced teachers create coherent lesson plans by aligning outcomes, classroom activities, and assessment strategies.	Alignment quality depends on the clarity of teacher-entered learning outcomes. The input page will guide teachers to enter clear and observable outcomes.
AI may replace teacher decision-making.	Use the Teacher Decision Service with Accept, Reject, Modify, and Annotate decision points.	Teachers must review recommendations first. Only accepted or modified recommendations can enter the final lesson plan, preserving teacher agency and professional judgement.	Some users may accept suggestions without reflection. Reflection prompts will ask users to justify why they accepted, rejected, or modified key recommendations.
LLM-generated lesson plans may be generic.	Use the Prompt Builder Service and Lesson Plan Generation Service with structured inputs: teacher context, analytics summary, teacher-approved	The LLM does not generate from a blank prompt or raw CSV alone. It generates a draft lesson plan grounded in data analysis, learning goals, and teacher-confirmed decisions.	LLM outputs may vary. The system will use fixed output formats, low-temperature settings, and an editable lesson plan editor.
Student data may create privacy risks.	Use synthetic or de-identified CSV/Excel data and avoid live school system integration. Raw CSV/XLSX files remain inside system storage and are not sent	The design supports privacy-preserving data use. The system does not require real student names or live connections to school systems. External AI receives only de-identified analytics summaries and teacher-approved decisions.	Users may accidentally upload identifiable data. The upload page will include privacy warnings and recommend anonymous student IDs such as S001 and S002.
Recommendation logic needs to be explainable and testable.	Use rule-based analytics and rule-based recommendation in the MVP.	Rule-based logic is transparent, easier to test, and easier to explain to teachers, which fits the Group B goal of making reasoning visible.	Rule-based logic is less flexible. Machine learning clustering or advanced analytics can be explored as future work.
Educators or researchers may need to understand how teachers interact with AI recommendations.	Use the Interaction Logging Service and Researcher Dashboard to record anonymised accept, reject, modify, and annotate patterns.	This helps educators or researchers observe how pre-service teachers understand and revise AI recommendations without exposing student identity.	The MVP dashboard may only show basic statistics. More detailed interaction analytics can be added later.

3.5.3 Alternative Solutions Considered

Considering the project objectives, project timeline, system feasibility, privacy risks, and the theoretical direction of Group B, the project team compared several feasible methods, as shown in the table below.

Design Decision	Alternative Options	Chosen Solution	Justification
Analytics method	Rule-based analytics vs machine learning clustering	Rule-based analytics	Rule-based analytics is more transparent, easier to test, and easier to explain to teachers. Since the project focuses on explainable teacher decision support rather than predictive modelling, it is more suitable for the MVP.
AI model strategy	Existing LLM API vs fine-tuned model	Existing LLM API	Using an existing LLM API is more feasible within the 10-week project timeframe and does not require a large dataset of teacher-reviewed lesson plans or pedagogical reasoning. Fine-tuning is treated as future work.
Lesson generation workflow	Direct AI lesson generation vs teacher-approved lesson generation	Teacher-approved lesson generation	Direct generation is faster but may replace teacher judgement. Teacher-approved generation keeps teacher agency and supports reflection before the final lesson plan is produced.
Knowledge retrieval	No RAG in MVP vs lightweight RAG	No RAG in MVP	The MVP prioritises the core workflow: upload, analytics, recommendation reasoning, teacher decision, lesson plan generation, and export. RAG adds knowledge-base maintenance and system complexity, so it is left as a future extension.
Data sharing with external AI	Send raw CSV to LLM vs send de-identified analytics summary only	Send de-identified analytics summary only	Raw CSV/XLSX files may contain sensitive or unnecessary details. Sending only analytics summaries, teacher context, and approved decisions reduces privacy risk and improves prompt controllability.

3.5.4 Novel Functionality Beyond Existing Systems

The project team studied two existing system categories: generative AI lesson planning tools and learning analytics dashboards. The limitations of these existing systems and the innovations of MentorPlan AI are shown in the table below.

Existing System Limitation	Our Novel Functionality
Generic generative AI lesson planning tools can quickly generate complete lesson plans, but they often do not explain why an activity, assessment, or differentiation strategy is selected.	The system provides recommendation plus expert reasoning explanation, allowing users to see the reasoning behind each teaching suggestion.
GenAI tools may encourage pre-service teachers to rely on complete AI-generated outputs instead of developing professional judgement.	The system requires teachers to accept, reject, modify, or annotate recommendations before final lesson plan generation.
Learning analytics dashboards can display student scores, participation, or quiz performance, but they often do not tell teachers what pedagogical action to take next.	The system converts learning analytics results into teaching recommendations such as grouping, reteaching, scaffolding, targeted practice, and extension.
Learning analytics dashboards rarely explain why a particular intervention, grouping strategy, or differentiation approach fits the student data.	Each recommendation includes data evidence and pedagogical rationale, showing how learning data influences the recommendation.
Existing systems often do not explain the relationship between learning outcomes, classroom activities, and assessment.	The Outcome-Activity-Assessment Alignment Explanation shows how recommended activities and assessments support stated learning outcomes.
Existing tools often focus on final output rather than teacher learning.	The system acts as an AI mentor that helps pre-service teachers observe, understand, and reflect on experienced teachers' planning logic.

3.5.5 Limitations and Mitigation Strategies

Although this MVP can support data analysis and expert reasoning, there are still some limitations, as shown in the table below.

Limitation	Potential Impact	Mitigation Strategy
LLM explanations may be generic or inaccurate.	Users may receive weak or partially inaccurate pedagogical explanations.	Use structured prompts requiring data evidence, learning outcome alignment, assessment alignment, and differentiation rationale. Teachers must review all recommendations.
Rule-based analytics may oversimplify student learning needs.	The system may miss complex learning patterns or hidden student groups.	Use transparent rules for the MVP to ensure explainability, and consider clustering or advanced learning analytics as future work.
Uploaded datasets may have inconsistent column names or formats.	The system may fail to parse datasets from different sources directly.	Provide data preview and a column mapping interface so users can map student_id, question_id, score, topic, skill_area, learning_outcome, and misconception_category fields.
Users may upload identifiable student data by mistake.	Privacy risks may occur.	Add upload warnings requiring synthetic or de-identified data and recommend anonymous student IDs.
External AI API is an external dependency.	The system may be affected by API availability, cost, latency, or output variability.	Use an LLM Gateway/Adapter, restrict data sent to AI, use fixed response schemas, and provide fallback messages or manual editing when needed.
Teachers may accept AI suggestions without reflection.	The system may become passive AI output consumption rather than a learning tool.	Add reflection prompts and require users to accept, reject, modify, or annotate key recommendations.
Generated lesson plans may not perfectly fit the classroom context.	The final lesson plan may still need teacher adjustment.	Provide an editable lesson plan editor and save revision history for teacher changes.
The MVP may not include a full teaching strategy knowledge base.	Recommendation coverage may be limited across teaching scenarios.	Start with predefined strategy categories and structured prompts. Add lightweight RAG or a strategy library as future work.
Researcher dashboard may only show basic interaction data.	The research value of early interaction analytics may be limited.	The MVP records accepted, rejected, modified, and annotated patterns first. More detailed analytics can be added later.
Privacy-preserving design may reduce dataset realism.	Synthetic or de-identified data may not fully represent classroom complexity.	Use diverse synthetic datasets to simulate different topics, score patterns, misconceptions, and learner needs. More realistic data can be considered later with appropriate approval.

Overall, the MVP design gives priority to explainability, feasibility, privacy protection, and teacher control. Complex models, full retrieval-augmented generation, and advanced learning analytics can be considered as future work. At present, the system focuses on clearly showing the relationship between student data, suggestions, expert reasoning and teachers' decisions.

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